

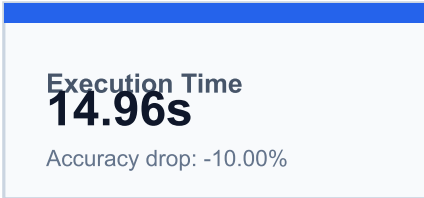
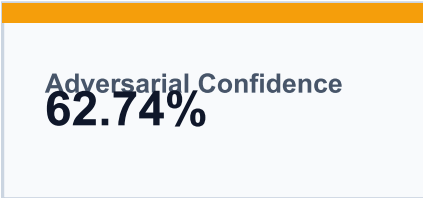
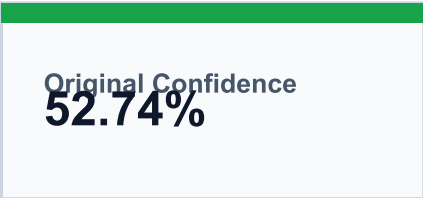
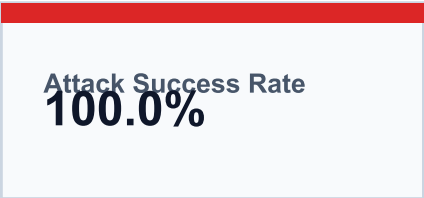
Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: google/efficientnet-b0

Attack Type: PGD

Images Evaluated: 9



Threat Level Assessment

HIGH RISK

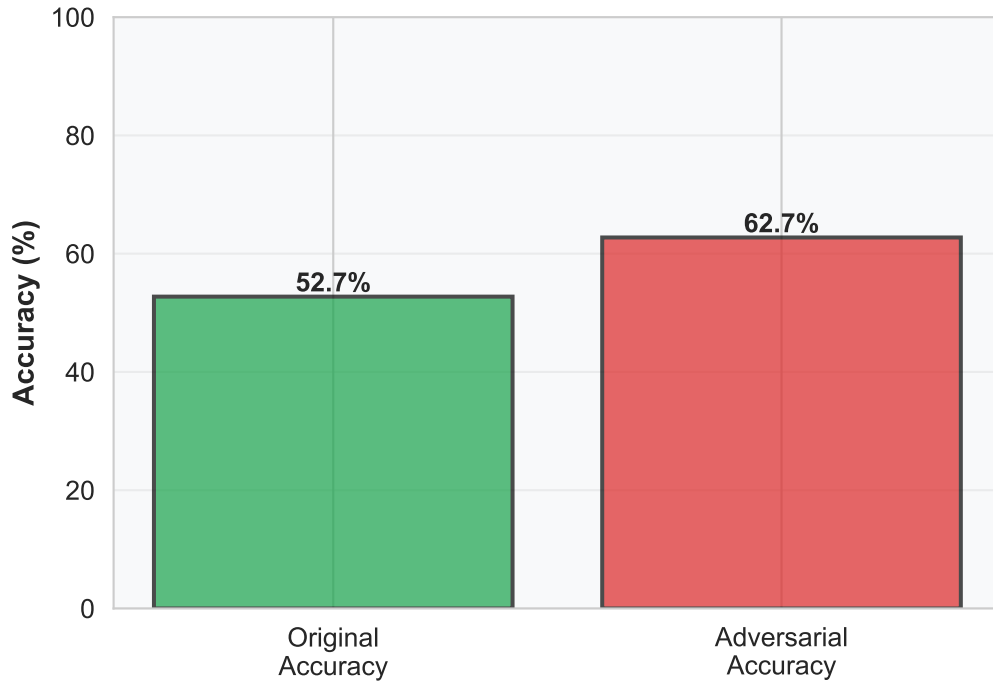
The PGD attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 52.74% to 62.74%, indicating the current robustness posture is high risk.

Assessment Narrative

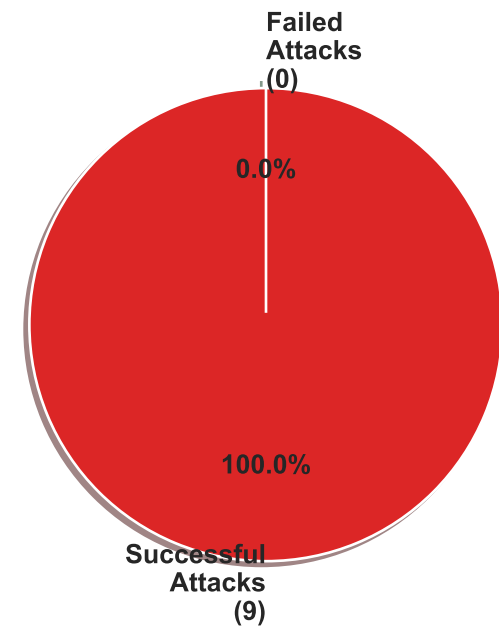
Successfully executed PGD attack on model google/efficientnet-b0 using 9 test images. Attack success rate: 100.0%. Average original accuracy: 52.74%, Average adversarial accuracy: 62.74%. The attack successfully fooled the model in 9 out of 9 cases.

Detailed Analysis

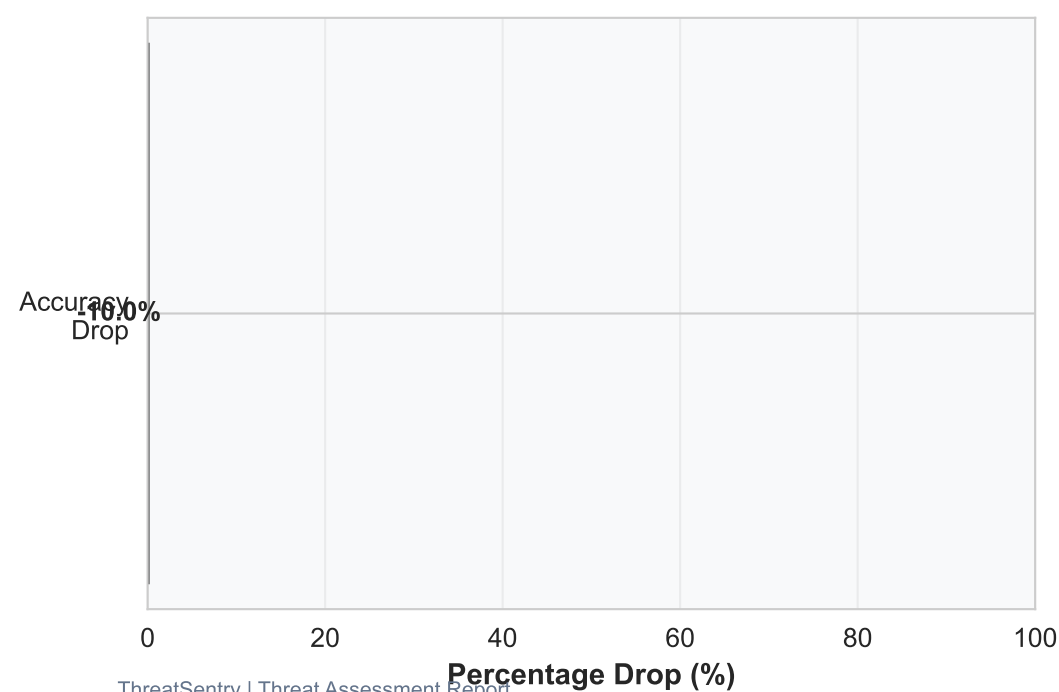
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

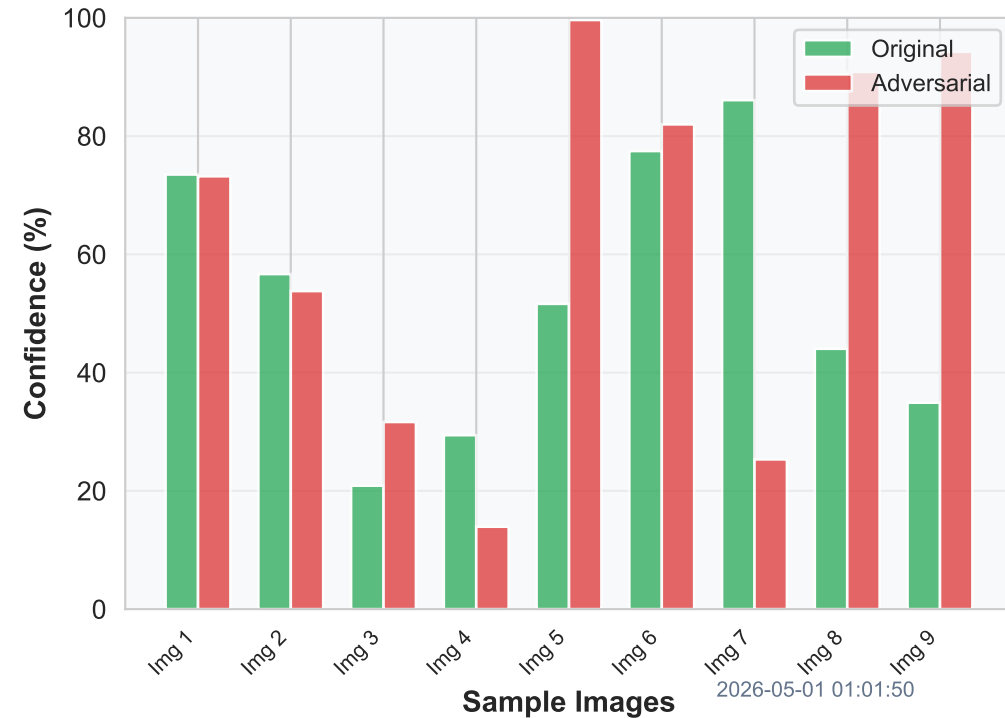


Image Evidence - Original vs Adversarial

Images 1-2 of 9

Original: titi, titi monkey
(51.6%)



Adversarial: three-toed sloth, ai, Bradypus tridactylus
(99.6%)
ATTACK SUCCESSFUL



Original: tiger, Panthera tigris
(73.5%)



Adversarial: zebra
(73.2%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 3-4 of 9

Original: hog, pig, grunter, squealer, *Sus scrofa*
(34.9%)



Adversarial: warthog
(94.2%)
ATTACK SUCCESSFUL



Original: coyote, prairie wolf, brush wolf, *Canis latrans*
(56.7%)



Adversarial: hog, pig, grunter, squealer, *Sus scrofa*
(53.8%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 5-6 of 9

Original: tusker
(44.0%)



Adversarial: African elephant, *Loxodonta africana*
(90.9%)
ATTACK SUCCESSFUL



Original: hartebeest
(29.4%)



Adversarial: lacewing, lacewing fly
(13.9%)
ATTACK SUCCESSFUL

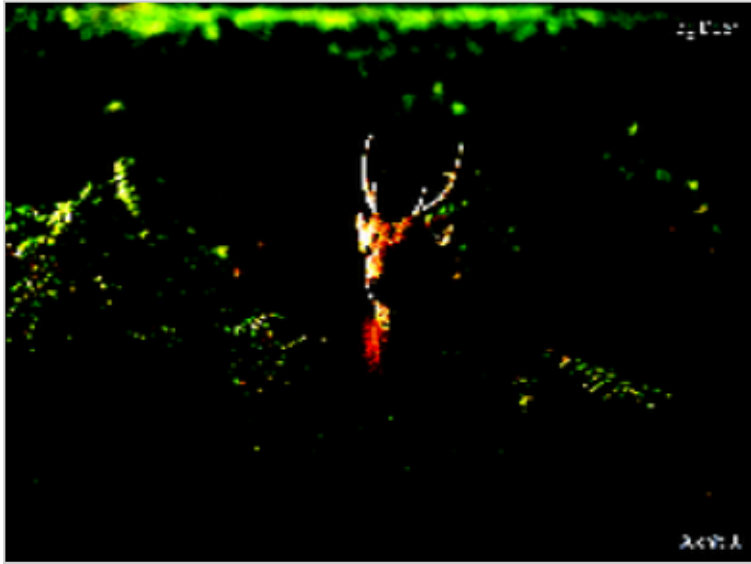


Image Evidence - Original vs Adversarial

Images 7-8 of 9

Original: indri, indris, Indri indri, Indri brevicaudatus
(20.9%)



Adversarial: macaque
(31.7%)
ATTACK SUCCESSFUL



Original: dhole, Cuon alpinus
(77.5%)



Adversarial: fountain
(82.0%)
ATTACK SUCCESSFUL

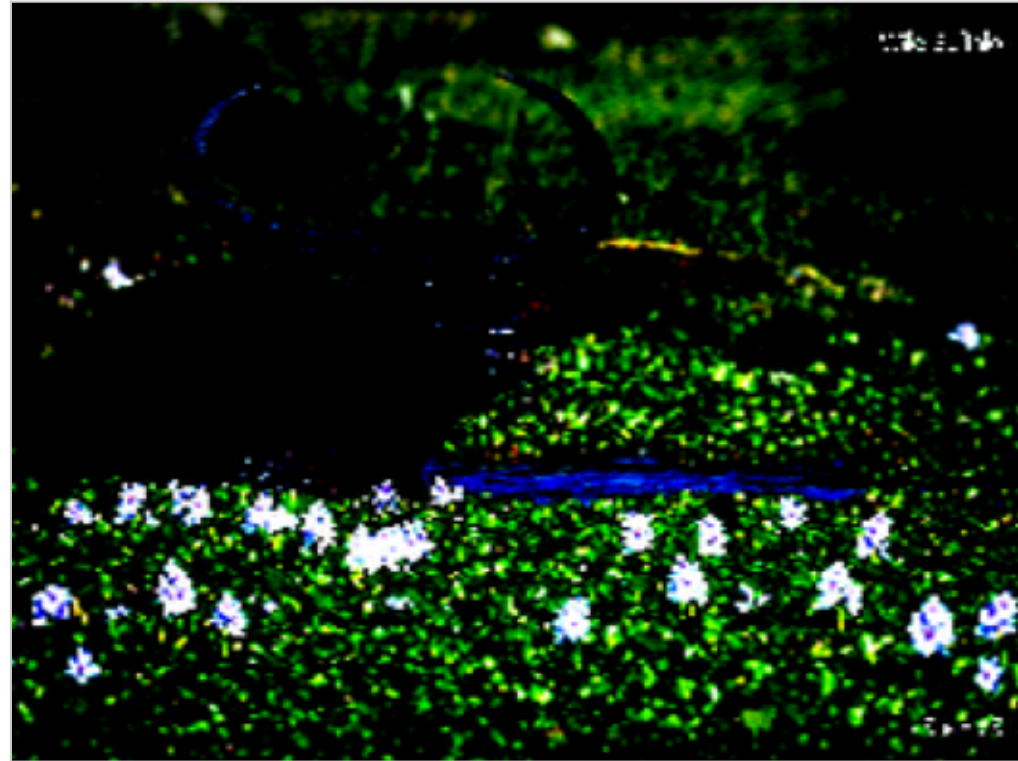


Image Evidence - Original vs Adversarial

Images 9-9 of 9

Original: water buffalo, water ox, Asiatic buffalo, Bubalus bubalis
(86.1%)

Adversarial: syringe
(25.3%)
ATTACK SUCCESSFUL



Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed PGD attack on model google/efficientnet-b0 using 9 test images. Attack success rate: 100.0%. Average original accuracy: 52.74%, Average adversarial accuracy: 62.74%. The attack successfully fooled the model in 9 out of 9 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques